

PORBIDO TRUCKING & HAULING SERVICE

Custom Transportation Management System (TMS) • Scope of Work & Feature Roadmap

Target Client: Porbido Management Client Partner: Cargill Philippines, Inc. Document Type: Technical Proposal (File 2 of 3)

1. Executive Summary & Core Project Objectives

The objective of this project is to build and deploy a modern, lightweight **Transportation Management System (TMS)** specifically engineered for **Porbido Trucking & Hauling Service**. The system replaces manual Excel spreadsheets, handwritten notes, and cluttered Messenger group chats with a streamlined digital pipeline.

Core System Mission

To eliminate data entry typos (weight tonnage, swapped destinations, bad TLOs), prevent unbilled or forgotten trips, automate freight rate math, and streamline driver cash settlements for the 2-person administrative team.

2. Phase 1: Core Operations & Error-Proof Billing System (MVP)

Focus: Eliminating immediate operational leaks, weight/route entry typos, and broken spreadsheet calculations.

Module 1: Smart Dispatch & TLO Validation Engine

PHASE 1 MVP

- **Dropdown Route Selectors:** Administrative staff choose pre-configured pickup and drop-off points (e.g., Subic → Pulilan, Pulilan → Iloilo, Cebu → Iloilo) from standardized dropdowns, eliminating destination mislabeling.
- **Strict TLO# Validation Rules:** Travel Load Order (TLO#) fields enforce strict numeric formatting, prevent duplicate entries, and mandate TLO presence before a trip can be saved.
- **Fleet & Driver Assignment Dropdowns:** Simple assignment controls linked to active 10-wheeler trucks (e.g., CCK 5273, NAK 2202, CAK 2693, CAO 3510) and registered drivers.

Module 2: Automated Freight & Rate Calculation Engine

PHASE 1 MVP

- **Automated Freight Math:** Completely replaces manual calculator work and prevents corrupted spreadsheet formulas (e.g., #REF! bugs).

$$\text{Freight Charge (₱)} = \text{Verified Weight (Tons)} \times \text{Route Market Rate (₱/Ton)}$$

- **Tonnage Sanity Validation:** Restricts weight entries to realistic operational bounds (e.g., 10.00 to 40.00 tons) and flags decimal transcription errors before saving.
- **Standardized Extra Charges Selector:** Pre-set input rules for fixed charges, re-route fees (e.g., P3,600.00 re-route), and demurrage fees.

Module 3: Driver Mobile POD & Receipt Queue

PHASE 1 MVP

- **Mobile Camera Upload Interface:** A lightweight, mobile-optimized page allowing drivers to upload crisp photos of signed Delivery Receipts (PODs) and weighbridge tickets directly linked to a specific Trip ID.
- **Admin Verification & Approval Queue:** Staff review uploaded receipt images and mark them as **"Approved"** or **"Flagged (Blurry/Illegible)"**, replacing endless searches through Messenger chat logs.

Module 4: Cargill TLO Billing Lifecycle Board

PHASE 1 MVP

- **Visual Trip Lifecycle Kanban:** Real-time visual tracking of every trip through clear operational stages:

Dispatched —► In Transit —► POD Uploaded —► Approved for Billing —► Billed to Cargill

- **Unbilled Trip Overdue Warnings:** Automated dashboard alert highlighting any completed trip stuck in "POD Uploaded" for over 48 hours to ensure zero revenue is forgotten.

3. Phase 2: Financial Reconciliation & Rolling Driver Ledger

Focus: Automated reconciliation against Cargill summaries, rolling driver balances, and 1-click billing exports.

Module 5: Rolling Driver Cash-on-Hand (COH) & Settlement Ledger

PHASE 2 EXPANSION

- **Automated COH Balance Engine:** Replaces paper liquidation notes with an automated rolling balance formula:

Ending COH = Starting COH + Trip Cash Advances – Validated Expenses (Fuel, Tolls, Meals)

- **Automatic Carry-Over:** The Ending COH balance of a finished trip automatically propagates as the Starting COH for that driver's next assigned haul.
- **Driver Salary & Helper Commission Rules:** Standardized payroll calculations deducting cash advances cleanly from net payouts.

Module 6: Cargill Statement Reconciliation Board

PHASE 2 EXPANSION

- **Summary Tally Matrix:** Cross-checks Porbido's internally logged trip data directly against Cargill's official 30-day billing summaries.
- **Discrepancy Highlighting:** Automatically flags any mismatch in tonnage, route rate, or TLO reference number between Porbido and Cargill records for immediate resolution before submitting statements.

Module 7: 1-Click Cargill PDF Statement Generator

PHASE 2 EXPANSION

- **Template-Matched PDF Export:** Generates fully formatted, audit-ready PDF billing summaries matching Cargill's exact submission layout (Summary 29 B – 50 B) with a single click.

4. Comprehensive System Feature Matrix

Feature Module	Key Business Value Delivered	Rollout Phase	Primary Users
Smart Dispatch & TLO Rules	Stops incorrect destination entries and bad/duplicate TLOs.	PHASE 1 MVP	Admin / Owner
Auto-Math Freight Engine	Eliminates weight typos, rate calculation errors, and broken Excel cells.	PHASE 1 MVP	Admin / Owner
Driver Mobile POD Queue	Centralizes Delivery Receipt photos; stops searching Messenger chat history.	PHASE 1 MVP	Drivers & Admin
Cargill TLO Billing Board	Ensures 100% trip billing recovery with overdue unbilled alerts.	PHASE 1 MVP	Admin / Owner
Rolling Driver COH Ledger	Automates cash advance carry-over balance and payroll deductions.	PHASE 2 EXPANSION	Admin / Owner
Cargill Reconciliation Matrix	Tallies internal records against Cargill summaries; catches underbillings.	PHASE 2 EXPANSION	Owner / Admin
1-Click PDF Statement Export	Saves days of manual spreadsheet creation for monthly billing.	PHASE 2 EXPANSION	Admin Staff

5. Technical Specifications & Architecture

- **Architecture:** Web-based Progressive Web Application (PWA) with responsive layouts optimized for desktop PCs (Admin) and mobile smartphones (Drivers).
- **Offline Capability:** Drivers can capture POD photos and submit status updates even in low-signal areas (quarries, highways); data auto-syncs when re-connected.
- **Security & Access Control:** Role-based access control separating Administrative Management views from Driver Submission portals.